

Summary

- ① CM5
- ② Overview
- ③ Programming by contract
 - Short view
 - The call-as-event idea
 - Main idea
- ④ Short Summary on Event-B
- ⑤ Internalizing events by merging
- ⑥ The Call-as-Event Paradigm
- ⑦ Analysis and then Synthesis
 - Analysis using Refinement
 - Synthesis by Merging
- ⑧ Summing a list of integer values

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Introduction

- Guidelines for using the transformations proposed by J.-R. Abrial and used for generating sequential algorithms.

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- Possibility to combine with others provers as TLAPS, Frama-c

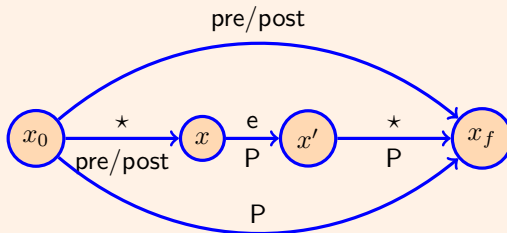
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- Illustrating the merging of events and the step to code.

- Guidelines for using the transformations proposed by J.-R. Abrial and used for generating sequential algorithms.
- Possibility to combine with others provers as TLAPS, Frama-c
- Illustrating the merging of events and the step to code.
- Relationship to PlusCal and TLA⁺ proof tools

Big steps versus Small steps

From pre/post to algorithm(s)

Refining a pre/post specification into a sequential algorithm



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Verification of contract (I)

A program P *satisfies* a contract $(\text{pre}, \text{post})$:

- P transforms a variable x from an initial value x_0 and produces a final value x_f : $x_0 \xrightarrow{P} x_f$
- x_0 satisfait pre : $\text{pre}(x_0)$ and x_f satisfait post : $\text{post}(x_0, x_f)$
- $\text{pre}(x_0) \wedge x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$

Floyd style

```
requires  $\text{pre}(x_0)$ 
ensures  $\text{post}(x_0, x_f)$ 
variables  $X$ 
[
  begin
  0 :  $P_0(x_0, x)$ 
  instruction0
  ...
  i :  $P_i(x_0, x)$ 
  ...
  instructionf-1
  f :  $P_f(x_0, x)$ 
  end
]
```

- $\text{pre}(x_0) \wedge x = x_0 \Rightarrow P_0(x_0, x)$
- $\text{pre}(x_0) \wedge P_f(x_0, x) \Rightarrow \text{post}(x_0, x)$
- For each pair ℓ, ℓ'
such that $\ell \longrightarrow \ell'$, one checks that
for any value $x, x' \in \text{MEMORY}$
$$\left(\begin{array}{l} \text{pre}(x_0) \wedge P_\ell(x_0, x) \\ \wedge \text{cond}_{\ell, \ell'}(x) \wedge x' = f_{\ell, \ell'}(x) \end{array} \right) \Rightarrow P_{\ell'}(x_0, x')$$

Verification du contract (II)

A program P *satisfies* a contract $(pre, post)$:

- P transforms a variable x from an initial value x_0 and produces a final value x_f : $x_0 \xrightarrow{P} x_f$
- x_0 satisfait pre : $pre(x_0)$ and x_f satisfait $post$: $post(x_0, x_f)$
- $pre(x_0) \wedge x_0 \xrightarrow{P} x_f \Rightarrow post(x_0, x_f)$

Hoare style

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- $\forall x_f, x_0. \text{pre}(x_0) \wedge x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$

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- $\forall x_f, x_0. \text{pre}(x_0) \Rightarrow (x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f))$

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- $\forall x_0. \text{pre}(x_0) \Rightarrow \forall x. (x_0 \xrightarrow{P} x \Rightarrow \text{post}(x_0, x))$
- $\forall x_0. \text{pre}(x_0) \Rightarrow WLP(P)(\text{post}(x_0, x))$

Listing 1 – Simple contract

```
#ifndef _A_H
#define _A_H
// Definition of the mathematical function mathpower2
/*@ axiomatic auxmath {
  @ lemma rule1: \forall int n; n > 0 ==> n*n == (n-1)*(n-1)+2*(n-1)+1;
  @ } */
/*@ requires 0 <= x;
  @ requires x*x <= INT_MAX;
  assigns \nothing;
  ensures \result == x*x;
*/
int power2(int x);
#endif
```

Listing 2 – Simple contract

```
#include <limits.h>
#include "power2.h"

int power2(int x)
{
    int r, k, cv, cw, or, ok, ocv, ocw;
    r=0; k=0; cv=0; cw=0; or=0; ok=k; ocv=cv; ocw=cw;
    /*@ loop invariant 0 <= cv && 0 <= cw && 0 <= k;
       @ loop invariant cv == k*k;
       @ loop invariant k <= x;
       @ loop invariant cw == 2*k;
       @ loop invariant 4*cv == cw*cw;
       @ loop assigns k, cv, cw, or, ok, ocv, ocw;
       @ loop variant x-k;
    */
    while (k<x)
    {
        ok=k; ocv=cv; ocw=cw;
        k=ok+1;
        cv=ocv+ocw+1;
        cw=ocw+2;
    }
    r=cv;
    return(r);
}
```

A Simple C Function

Listing 3 – Simple contract

```
frama-c -wp -rte power2.c
[kernel] Parsing power2.c (with preprocessing)
[rte:annot] annotating function power2
[wp] 30 goals scheduled
[wp] Proved goals: 32 / 32
Terminating: 1
Unreachable: 1
Qed: 22 (5ms-1ms-7ms)
Alt-Ergo 2.6.0: 8 (13ms-15ms-18ms)
```

- Finding or computing annotations is difficult and even undecidable!

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- Given

Contract EX
 Variables $X(int)$
 Requires $pre(x_0)$
 Ensures $post(x_0, x_f)$

Correct by Construction

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- Design of the algorithm ALG fulfilling the contract

- Finding or computing annotations is difficult and even undecidable !

- Given

Contract EX Variables $X(int)$ Requires $pre(x_0)$ Ensures $post(x_0, x_f)$
--

- Design of the algorithm ALG fulfilling the contract
- HOARE triple :

$$\{\mathbf{pre}(\mathbf{x}_0) \wedge \mathbf{x} = \mathbf{x}_0\} \text{ALG} \{\mathbf{post}(\mathbf{x}_0, \mathbf{x})\}$$

- Finding or computing annotations is difficult and even undecidable!

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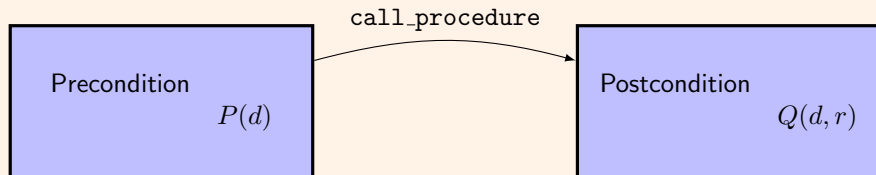
Idea ?

From Contract *EX*, using a step by step approach to find an algorithm
ALG satisfying it using refinement and Event-B models.
Applying the CLEANROOM model but with tool support

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- Providing patterns for helping the use of refinement.
- Making proofs as simple as possible.
- Simplifying the discovery of invariants
- we focus on the transformation of the final concrete specification into a PlusCal algorithm :
 - ① transforming an `EVENT B` specification into a PlusCal algorithm
 - ② using the merging of events to get only one event corresponding to the call of the algorithm
 - ③ double checking the resulting algorithm by using TLA^+ and its proof system.

The call as event idea



- `call_procedure` is simulated by the event `call_procedure` :

```
EVENT CALL_PROCEDURE  
WHEN  
   $P(d)$   
THEN  
   $r : Q(d, r')$   
END
```

- $P(d)$ is specifying the inputs of the procedure.
- $Q(d, r)$ is expressing the relation between the inputs d and the results r .

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- A pre/post specification is a pair of assertions $(pre, post)$
- $(pre, post)$ is defining a *transformation* over variables that is correct with respect to *HOARE* triples :

$$\{pre\} transformation \{post\}$$

- **Problem** : Design of a program PROG simulating the *transformation* with respect to $(pre, post)$:

$$\{pre\} PROG \{post\}$$

- **Idea** : Writing a *one-shot* relation simulating the relation over *flexible* variables :

$$\forall x, y. pre(x) \wedge R(x, y) \Rightarrow post(x, y)$$

- **event** : Expressing the relation :

```
EVENT  PROG
WHEN
  pre(x)
THEN
  y : |post(x, y')
END
```

The sequential pattern

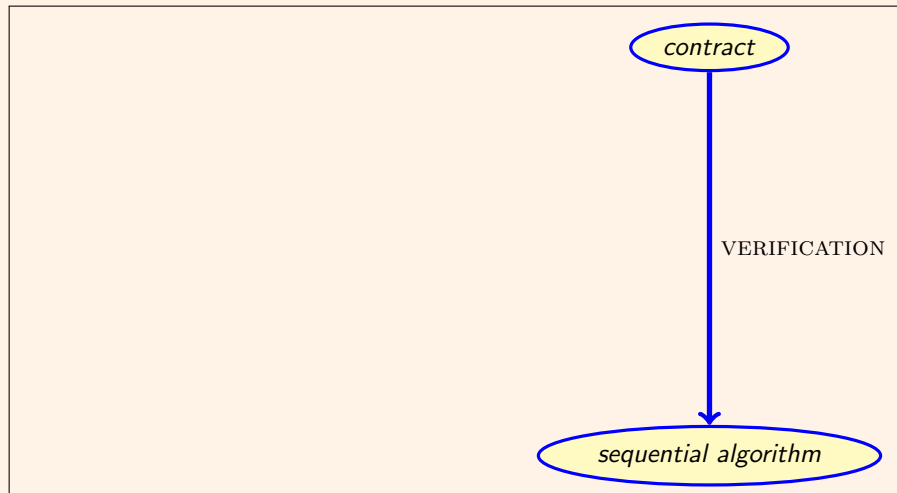
contract

The sequential pattern

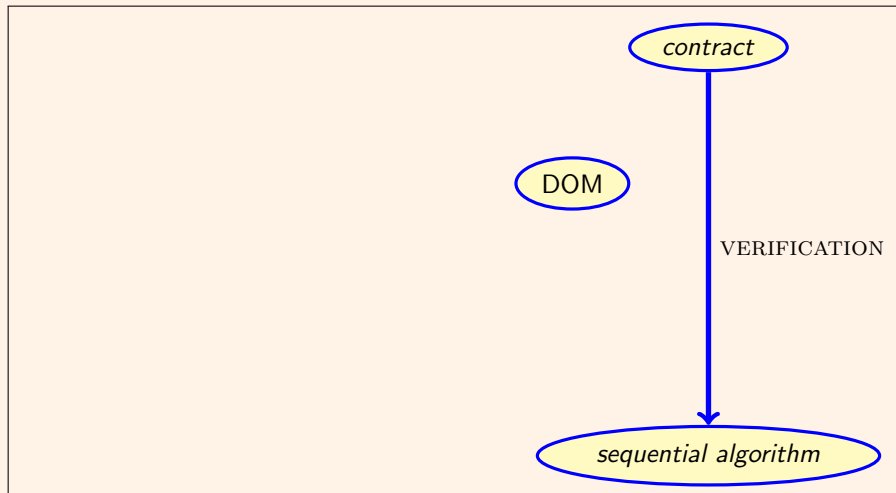
contract

sequential algorithm

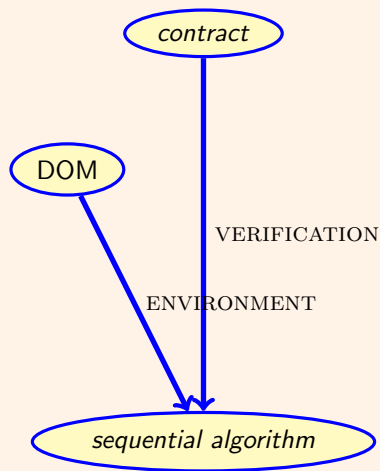
The sequential pattern



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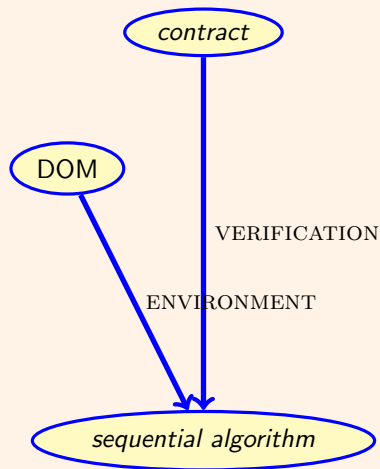


The sequential pattern

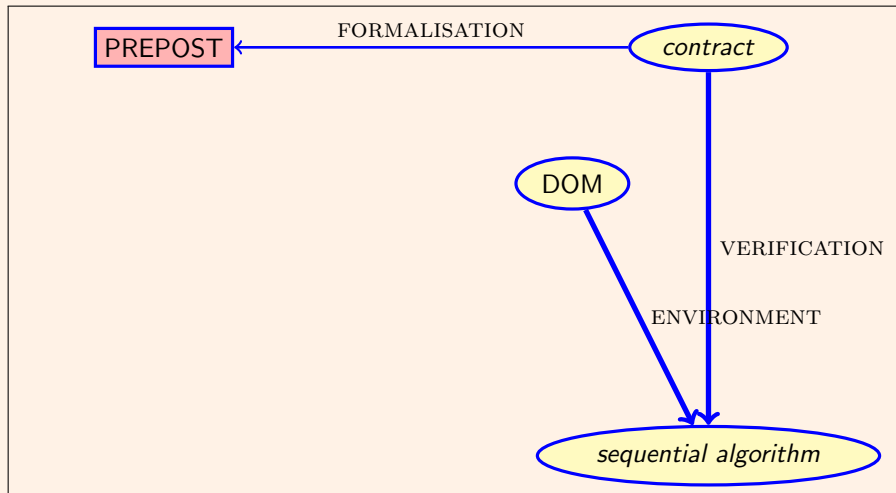


The sequential pattern

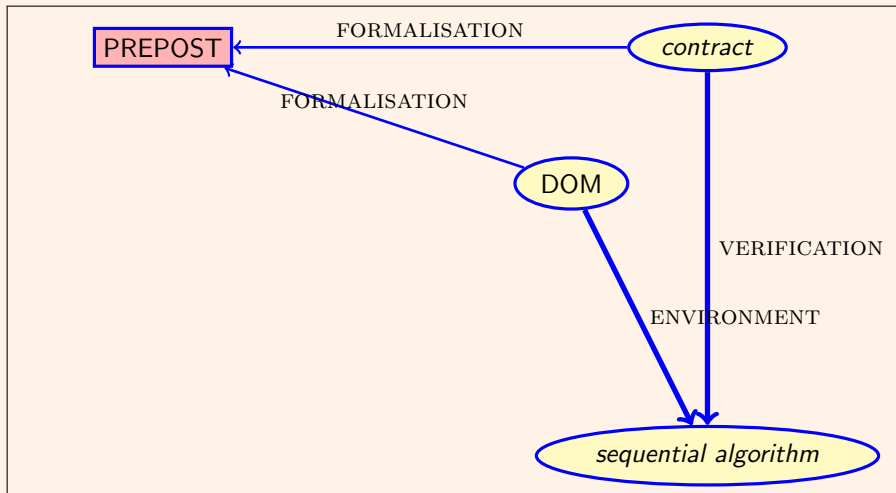
PREPOST



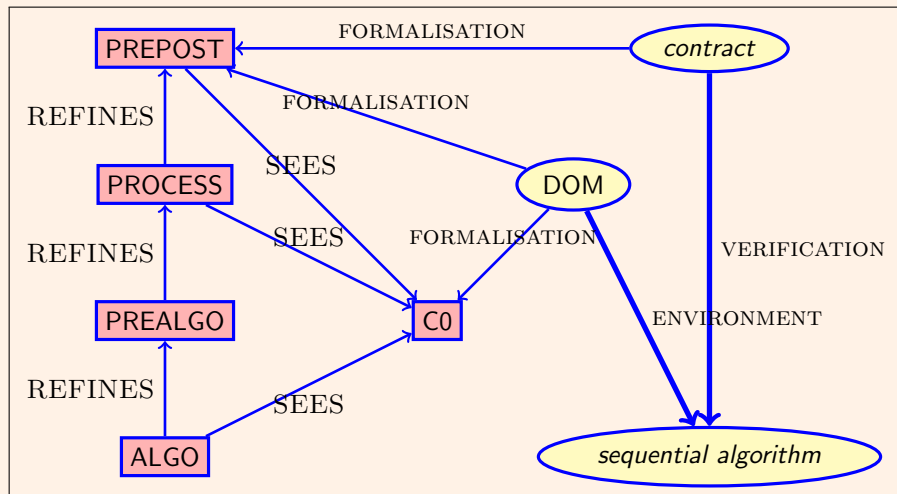
The sequential pattern



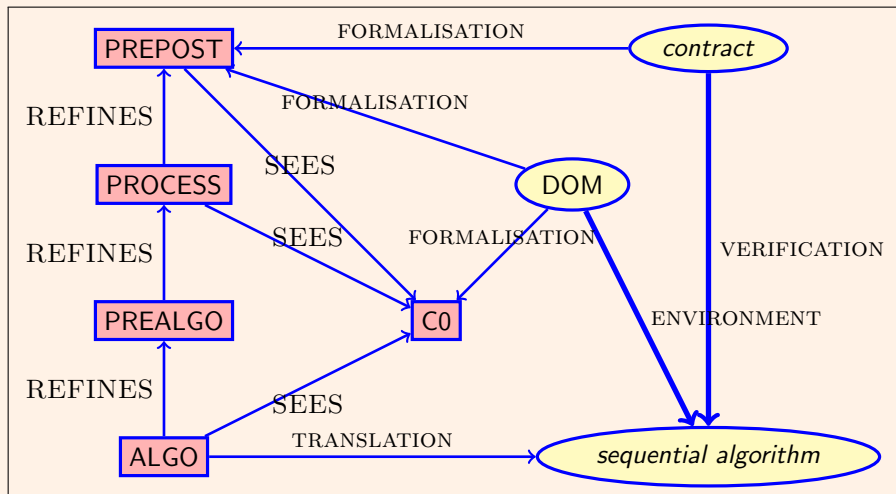
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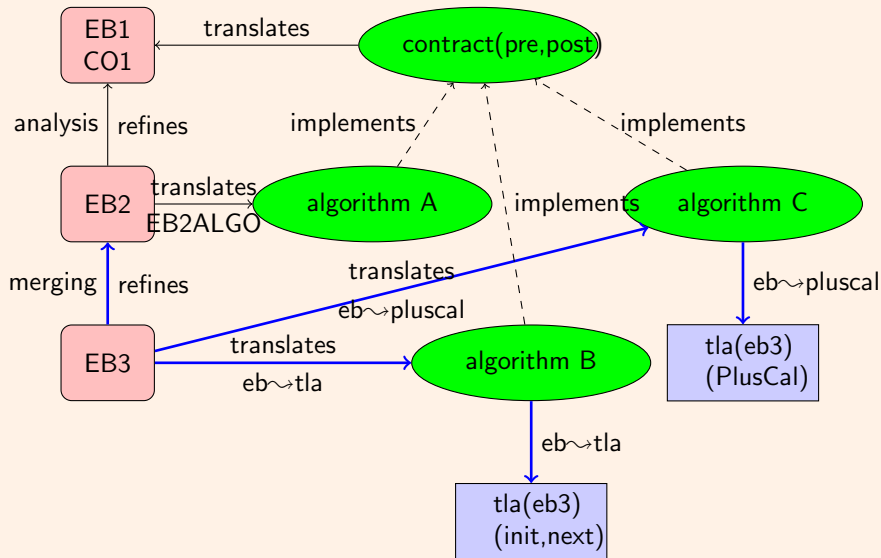
The sequential pattern



The sequential pattern



Global Picture



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Short Summary on Event-B (context)

- Context : static properties of Event-B models
 - ▶ Sets : user-defined types
 - ▶ Constants : static object in development
 - ▶ Axioms : presumed properties about sets and constants
 - ▶ Theorems : derived properties about sets and constants

SETS

A

CONSTANTS

B, C, f

AXIOMS

$ax1 : B \subseteq A$

$ax2 : C \subseteq A$

$ax3 : g \in B \rightarrow C$

$ax4 : \forall A. A \subseteq \mathbb{N} \wedge 0 \in A \wedge suc[A] \subseteq A \Rightarrow \mathbb{N} \subseteq A$

...

Short Summary on Event-B (discrete)

- Machine : behavioral properties of Event-B models
 - Variables : states
 - Invariants : properties of variables that always need to hold
 - Theorems : derived properties about variables
 - Events : possible state changes
 - c et s are constantes and visible sets by e
 - x is a state variable or a list of variables
 - $G(c, s, t, x)$ is the condition for observing e .
 - $P(c, s, t, x, x')$ is the assertion for the relation over x and x' .
 - $BA(e)(c, s, x, x')$ is the *before-after* relationship for e and is defined by $\exists t. G(c, s, t, x) \wedge P(c, s, t, x, x')$.
- ```

EVENT e
ANY t
WHERE
 G(c, s, t, x)
THEN
 x : |(P(c, s, t, x, x'))
END

```

```

EVENT e
ANY t
WHERE
 $G(c, s, t, x)$
THEN
 $x : |(P(c, s, t, x, x'))|$
END

```

# Short Summary on Event-B (refinement)

- Given an **abstract** and a corresponding **concrete** event

**EVENT**  $ae \hat{=}$   
**ANY**  $v$  **WHERE**  
 $G(x, v)$   
**THEN**  
 $x := E(x, v)$   
**END**

**EVENT**  $ce \hat{=}$   
**ANY**  $w$  **WHERE**  
 $H(y, w)$   
**THEN**  
 $y := F(y, w)$   
**END**

$$\begin{aligned} I(x) \wedge J(x, y) \wedge H(y, w) \\ \implies \\ \exists v \cdot (G(x, v) \wedge J(E(x, v), F(y, w))) \end{aligned}$$

- $BA(ae)(x, x') \hat{=} \exists v. G(x, v) \wedge x' = E(x)$
- $BA(ce)(y, y') \hat{=} \exists w. H(y, w) \wedge y' = F(y)$

# Modelling systems in Event-B

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**MACHINE**

*m*

**SEES**

*c*

**VARIABLES**

*x*

**INVARIANT**

*I(x)*

**THEOREMS**

*Q(x)*

**INITIALISATION**

*Init(x)*

**EVENTS**

*...e*

**END**

# Modelling systems in Event-B

$c$  defines the static environment for the proofs  
related to  $m$  : sets, constants, axioms, theorems  $\Gamma(m)$ .

**MACHINE**

$m$

**SEES**

$c$

**VARIABLES**

$x$

**INVARIANT**

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**EVENTS**

$\dots e$

**END**

# Modelling systems in Event-B

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**INITIALISATION**

$Init(x)$

**EVENTS**

$\dots e$

**END**

$c$  defines the static environment for the proofs related to  $m$  : sets, constants, axioms, theorems  $\Gamma(m)$ .

$$\Gamma(m) \vdash \forall x \in Values : \text{INIT}(x) \Rightarrow I(x)$$

# Modelling systems in Event-B

**MACHINE**

*m*

**SEES**

*c*

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*x*

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*I(x)*

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*Init(x)*

**EVENTS**

*... e*

**END**

*c* defines the static environment for the proofs related to *m* : sets, constants, axioms, theorems  $\Gamma(m)$ .

$\Gamma(m) \vdash \forall x \in Values : \text{INIT}(x) \Rightarrow I(x)$

$\forall e :$

$\Gamma(m) \vdash \forall x, x', u \in Values : I(x) \wedge R(u, x, x') \Rightarrow I(x')$

# Modelling systems in Event-B

## MACHINE

 $m$ 

## SEES

*C*

## VARIABLES

 $\mathcal{X}$ 

## INVARIANT

$$I(x)$$

## THEOREMS

$$Q(x)$$

## INITIALISATION

$$Init(x)$$

## EVENTS

 $\dots e$ 

END

**c** defines the static environment for the proofs  
related to  $m$  : sets, constants, axioms, theorems  $\Gamma(m)$ .

$$\Gamma(m) \vdash \forall x \in Values : \text{INIT}(x) \Rightarrow \text{I}(x)$$
 $\forall e:$ 
$$\Gamma(m) \vdash \forall x, x', u \in Values : I(x) \wedge R(u, x, x') \Rightarrow I(x')$$
$$\Gamma(m) \vdash \forall x \in Values : I(x) \Rightarrow Q(x)$$

# Modelling systems in Event-B

**MACHINE**

*m*

**SEES**

*c*

**VARIABLES**

*x*

**INVARIANT**

*I(x)*

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*Q(x)*

**INITIALISATION**

*Init(x)*

**EVENTS**

*... e*

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$\Gamma(m) \vdash \forall x \in Values : \text{INIT}(x) \Rightarrow I(x)$

$\forall e :$

$\Gamma(m) \vdash \forall x, x', u \in Values : I(x) \wedge R(u, x, x') \Rightarrow I(x')$

$\Gamma(m) \vdash \forall x \in Values : I(x) \Rightarrow Q(x)$

*e*

**ANY**

*u*

**WHERE**

*G(x, u)*

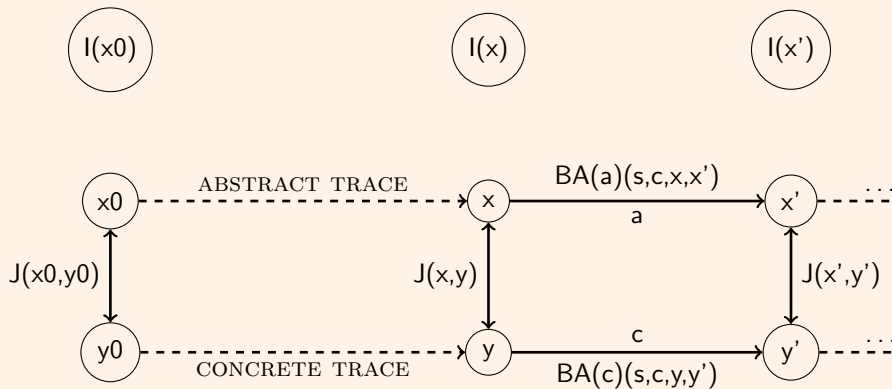
**THEN**

*x : |(R(u, x, x'))*

**END**

or *e* is **observed**  $x \xrightarrow{e} x'$

# Refinement between two machines



## Current Section Summary

- 1 CM5
- 2 Overview
- 3 Programming by contract
  - Short view
  - The call-as-event idea
  - Main idea
- 4 Short Summary on Event-B
- 5 Internalizing events by merging
- 6 The Call-as-Event Paradigm
- 7 Analysis and then Synthesis
  - Analysis using Refinement
  - Synthesis by Merging
- 8 Summing a list of integer values

# Merging two events

**EVENT** ae1

**ANY**  $u$  **WHERE**

$G1(u, s, c, x)$

**THEN**

$x : |ABAP(u, s, c, x, x')$

**END**

**EVENT** ae2

**ANY**  $u$  **WHERE**

$G2(u, s, c, x)$

**THEN**

$x : |ABAP(u, s, c, x, x')$

**END**

refines

refines

**EVENT** c

**REFINES** ae1, ae2

**ANY**  $u$  **WHERE**

$H(u, s, c, u)$

**THEN**

$x : |ABAP(u, s, c, x, x')$

**END**

# Merging two events

**EVENT** ae1

**ANY**  $u$  **WHERE**

$G1(u, s, c, x)$

**THEN**

$x : |ABAP(u, s, c, x, x')$

**END**

**EVENT** ae2

**ANY**  $u$  **WHERE**

$G2(u, s, c, x)$

**THEN**

$x : |ABAP(u, s, c, x, x')$

**END**

refines

refines

**EVENT** c

**REFINES** ae1, ae2

**ANY**  $u$  **WHERE**

$H(u, s, c, u)$

**THEN**

$x : |ABAP(u, s, c, x, x')$

**END**

$AX(s, c), I(s, c, x), H(v, s, c, y) \vdash G_1(s, c, x) \vee G_2(s, c, x).$

# Applying the fusion of events

## Strategy for merging events

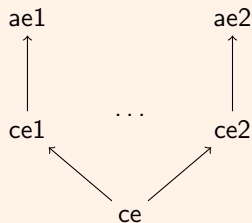
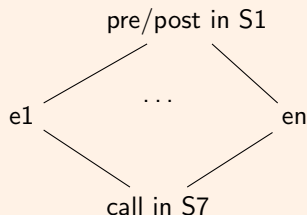
Let be two events  $ae1$  and  $ae2$  such that the two guards are complementary (ie  $g(ae1) \Leftrightarrow \neg g(ae2)$  or  $g(ae2) \Leftrightarrow \neg g(ae1)$  ).

- Let  $cei$  be the event with the guard  $g(aei)$  and with the action  $(g(ae1) \Rightarrow a(ae1)) \wedge (g(ae2) \Rightarrow a(ae2))$ .

**Then  $cei$  refines  $aei$ .**

- Let  $ce$  be the event with the guard  $g(ae1) \vee g(ae2)$ .

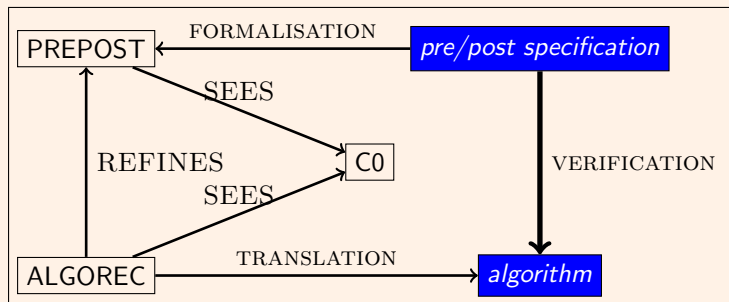
**Then  $ce$  refines  $cei$  by fusion.**



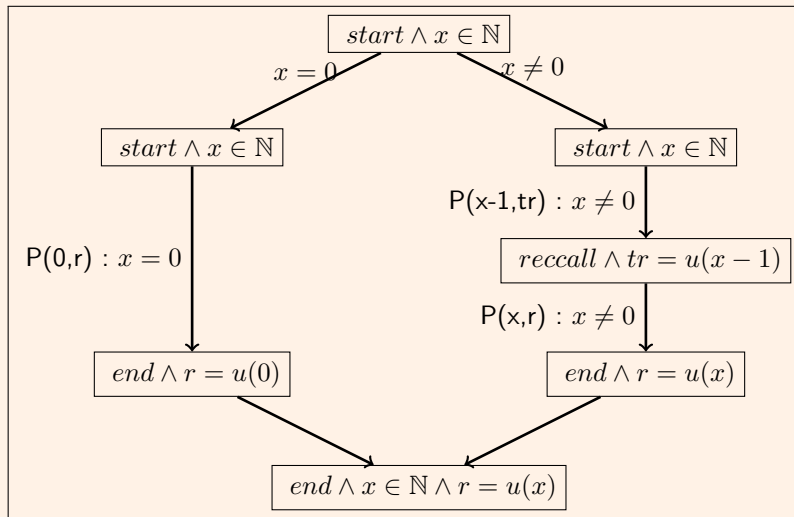
## Current Section Summary

- ① CM5
- ② Overview
- ③ Programming by contract
  - Short view
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# The recursive pattern



# The refinement diagram



# Ideas for producing algorithms

- **EVENT**  $\text{rec}\% \text{PROC}(h(x), y) \% P(y)$  is simply simulating the recursive call of the same function.
- The invariant is defined in a simpler way by analysing the inductive structure and a control variable is introduced for structuring the inductive computation.

```
EVENT
e
WHERE
 $\ell = \ell_1$
 $g_{\ell_1, \ell_2}(x)$
THEN
 $\ell := \ell_2$
 $x := f_{\ell_1, \ell_2}(x)$
END
```

```
EVENT
rec%PROC(h(x),y)%P(y)
ANY y
WHERE
 $\ell = \ell_1$
 $g_{\ell_1, \ell_2}(x, y)$
THEN
 $\ell := \ell_2$
 $x := f_{\ell_1, \ell_2}(x, y)$
END
```

```
EVENT
call%APROC(h(x),y)%P(y)
ANY y
WHERE
 $\ell = \ell_1$
 $g_{\ell_1, \ell_2}(x, y)$
THEN
 $\ell := \ell_2$
 $x := f_{\ell_1, \ell_2}(x, y)$
END
```

# The computation of the function $x^2$

## VARIABLES

$r, c, tr$

## INVARIANTS

$inv1 : r \in \mathbb{N}$

$inv2 : c = end \Rightarrow r = n * n$

$inv3 : c = callrec \Rightarrow n \neq 0$

$inv4 : c = callrec \Rightarrow tr = (n - 1) * (n - 1)$

$inv5 : c \in C$

$inv6 : tr \in \mathbb{N}$

$inv7 : c = end \Rightarrow r = n * n$

$inv8 : c = end \wedge n \neq 0$

$\Rightarrow tr = (n - 1) * (n - 1) \wedge r = tr + 2 * (n - 1) + 1$

$inv9 : c = callrec \Rightarrow n * n = tr + 2 * (n - 1) + 1$

# The computation of the function $x^2$

```

EVENT INITIALISATION
BEGIN
 act1 : r := 0
 act2 : c := start
 act3 : tr ∈ ℕ
END
EVENT square0
REFINES square(n; r)
WHEN
 grd1 : c = start
 grd2 : n = 0
THEN
 act1 : c := end
 act2 : r := 0
END

```

```

EVENT squaren
REFINES square(n ; r)
 WHEN
 $grd1 : c = callrec$
 THEN
 $act1 : r := tr + 2 * (n - 1) + 1$
 $act2 : c := end$
 END
EVENT rec%square($n-1$; tr)
 WHEN
 $grd1 : c = start$
 $grd2 : n \neq 0$
 THEN
 $act1 : c := callrec$
 $act2 : tr := (n - 1) * (n - 1)$
 END

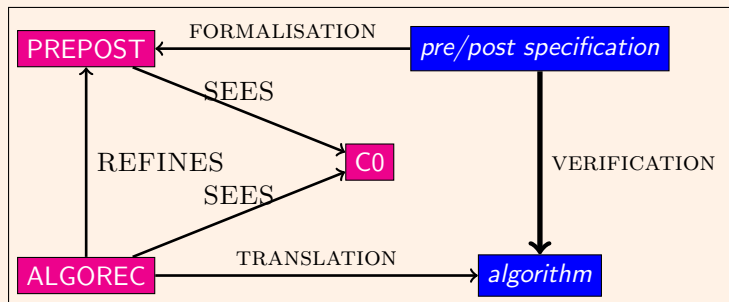
```

## Summary for proof obligations

## Summary for proof obligations

| Name       | Total | Automatic | Interactive |
|------------|-------|-----------|-------------|
| cae-square | 34    | 32        | 2           |
| square0    | 3     | 2         | 1           |
| specsquare | 2     | 2         | 0           |
| square     | 29    | 28        | 1           |

# The recursive pattern



- Proofs are easier and simpler
- Invariant is simple to find
- Translation is automatic
- Students are not happy with it and tools are not always set for this verification.

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Analysis using Refinement  
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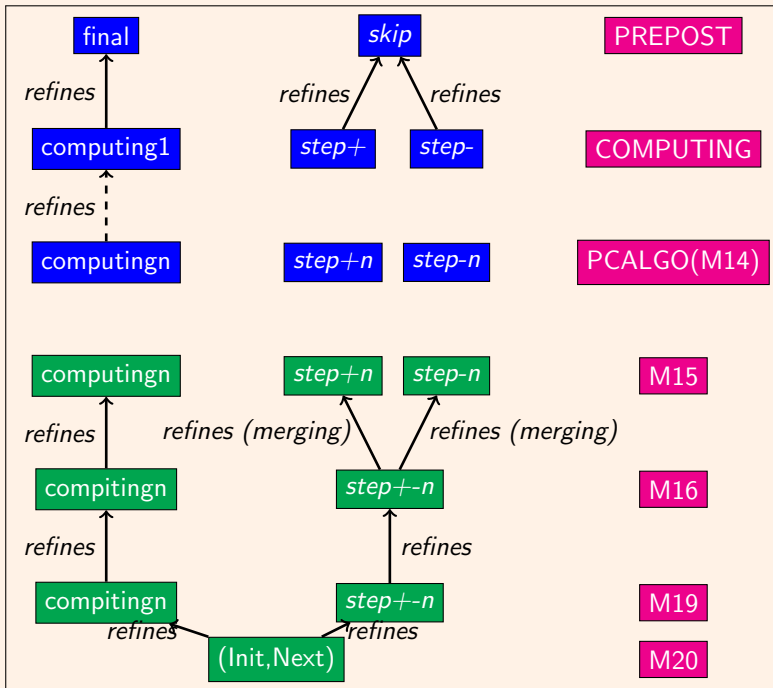
## First n numbers and first odd/even numbers

- First the **pre/post specification** ...
- Possibility to use a programming language with contracts too

- ### h contracts too

the sum of first  $n$   
 numbers and first even  
 smaller than  $n$   
 :

- sequences  $s$  and  $es$   
 the sum of first  $n$   
 numbers and first even  
 smaller than  $n$   
 :  
 $\mathbb{N} \Rightarrow s(n) = \sum_{i=0}^{i=n} i.$   
 $\mathbb{N} \Rightarrow es(n) =$   
 $2 * i.$



## First n numbers and first odd/even numbers

$$axm1 : n \in \mathbb{N}$$

$$axm2 : s \in \mathbb{N} \rightarrow \mathbb{N} \wedge os \in \mathbb{N} \rightarrow \mathbb{N} \wedge es \in \mathbb{N} \rightarrow \mathbb{N}$$

$$axm3 : es(0) = 0 \wedge os(0) = 0 \wedge s(0) = 0$$

$$axm4 : \forall i, l. i \in \mathbb{N} \wedge l \in \mathbb{N} \wedge i = 2 * l \Rightarrow s(i+1) = s(i) + i + 1 \wedge es(i+1) =$$

$$axm5 : \forall i, l. i \in \mathbb{N} \wedge l \in \mathbb{N} \wedge i = 2 * l + 1 \Rightarrow s(i+1) = s(i) + i + 1 \wedge es(i+1) =$$

$$axm6 : suc \in \mathbb{N} \rightarrow \mathbb{N} \wedge (\forall i. i \in \mathbb{N} \Rightarrow suc(i) = i + 1)$$

$$axm7 : \forall A. A \subseteq \mathbb{N} \wedge 0 \in A \wedge suc[A] \subseteq A \Rightarrow \mathbb{N} \subseteq A$$

$$th1 : \forall i. i \in \mathbb{N} \Rightarrow s(i+1) = s(i) + i + 1$$

$$th2 : \forall u, v. u \in \mathbb{N} \wedge v \in \mathbb{N} \wedge 2 * u = v \Rightarrow u = v/2$$

$$th3 : \forall k. k \in \mathbb{N} \Rightarrow 2 * s(k) = k * k + k$$

$$th4 : \forall k. k \in \mathbb{N} \Rightarrow s(k) = (k * k + k)/2$$

$$th5 : \forall k. k \in \mathbb{N} \Rightarrow es(2 * k) = 2 * s(k)$$

$$th6 : \forall k. k \in \mathbb{N} \Rightarrow es(2 * k + 1) = 2 * s(k)$$

$$th7 : \forall k. k \in \mathbb{N} \wedge k \neq 0 \Rightarrow os(2 * k) = k * k$$

$$th8 : \forall k. k \in \mathbb{N} \Rightarrow os(2 * k + 1) = (k + 1) * (k + 1)$$

## INVARIANTS

$$inv1 : input \in \mathbb{Z}$$
$$inv6 : input = n$$
$$inv2 : rs \in \mathbb{Z} \wedge re \in \mathbb{Z}$$
$$inv3 : ok \in \mathit{BOOL}$$
$$inv4 : ok = TRUE \Rightarrow rs = s(input) \wedge re = es(input)$$

END

## Stating the pre/post specification in Event-B

## Refining for computing (1)

## INVARIANTS

$$inv1 : cur \in 0 .. n$$
$$inv2 : ee \in 0..n \mapsto \mathbb{N}$$
$$inv3 : ss \in 0..n \mapsto \mathbb{N}$$
$$inv5 : dom(ss) = 0..cur \wedge dom(ee) = dom(ss) \wedge dom(ss) \subseteq \mathbb{N}$$
$$inv6 : \forall i. i \in 0 .. cur \Rightarrow ee(i) = es(i) \wedge ss(i) = s(i)$$

## VARIANT

## THEN

$$act1 : ok := FALSE$$
$$act2 : rs : \in \mathbb{Z}$$
$$act3 : re : \in \mathbb{Z}$$
$$act4 : input := n$$
$$act5 : ee := \{0 \mapsto 0\}$$
$$act6 : ss := \{0 \mapsto 0\}$$
$$act7 : cur := 0$$

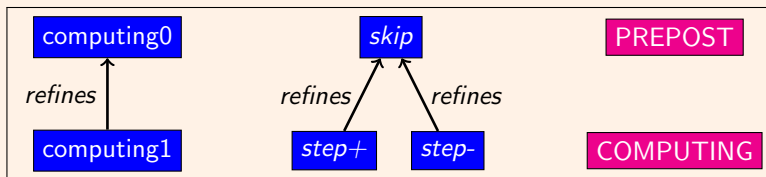
END

## Refining for computing (2)

## Refining for computing (3)

## Refining for computing (4)

# Diagram of events



## Completing the analysis

**MACHINE  $M_1$  SEES  $C_0$**

## VARIABLES

 $input, rs, re, ok,$ 

## INVARIANTS

$$inv1 : input \in \mathbb{Z}$$
$$inv6 : input = n$$
$$inv2 : rs \in \mathbb{Z} \wedge re \in \mathbb{Z}$$
$$inv3 : ok \in \mathit{BOOL}$$
$$inv4 : ok = TRUE \Rightarrow rs = s(input) \wedge re = es(input)$$

## Completing the analysis

MACHINE  $M_{11}$  SEES  $C_0$

## VARIABLES

 $input, rs, re, ok, ee, ss, cur,$ 

## INVARIANTS

$$inv1 : cur \in 0..n$$
$$inv2 : ee \in 0 .. n \mapsto \mathbb{N}$$
$$inv3 : ss \in 0 .. n \mapsto \mathbb{N}$$
$$inv5 : dom(ss) = 0 .. cur \wedge dom(ee) = dom(ss) \wedge dom(ss) \subseteq \mathbb{N}$$
$$inv6 : \forall i. i \in 0 .. cur \Rightarrow ee(i) = es(i) \wedge ss(i) = s(i)$$

## Completing the analysis

**MACHINE**  $M_{12}$  **SEES**  $C_0$

## VARIABLES

 $input, rs, re, ok, ee, ss, cur, cs, ce,$ 

## INVARIANTS

$$inv1 : cs = ss(cur)$$
$$inv2 : ce = ee(cur)$$

## Completing the analysis

MACHINE  $M_{13}$  SEES  $C_0$

## VARIABLES

 $input, rs, re, ok, ee, ss, cur, cs, ce,$ 

## INVARIANTS

$$inv1 : cs = ss(cur)$$
$$inv2 : ce = ee(cur)$$
$$inv4 : ce = es(cur)$$
$$inv3 : cs = s(cur)$$

## Completing the analysis

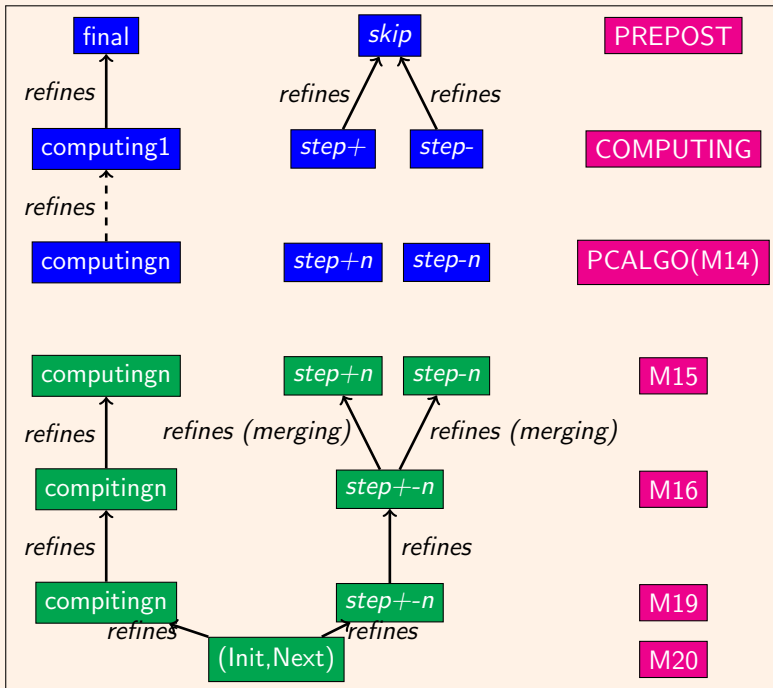
**MACHINE**  $M_{14}$  **SEES**  $C_0, C_1$

## VARIABLES

 $input, rs, re, cur, cs, ce, l,$ 

## INVARIANTS

$$inv1 : cs = s(cur)$$
$$inv5 : l = start \Leftrightarrow ok = FALSE$$
$$inv6 : l = end \Leftrightarrow ok = TRUE$$
$$inv2 : cs = s(cur)$$
$$inv3 : l \in L$$
$$inv4 : l = end \Rightarrow re = es(n) \wedge rs = s(n)$$



## Translation from Events

```
struct sums codesum(int n)
{int k,ce,cs; struct sums r;
 r.s=0;r.se=0;k=0;ce=0;cs=0;
 while (k<n)
 {
 if (k % 2 != 0)
 { ce = ce + k + 1;cs = cs +k+1; k = k +1;}
 else
 { ce = ce ;cs = cs +k+1; k = k +1;}
 }
 r.s=cs;r.se=ce;
 return(r);
}
```

# Generation of an algorithm from M14

## Improving and checking with Frama-c

```
#include "structure.h"
#include "trans-even.h"
struct sums codesum(int n)
{
 int k, ce, cs; struct sums r;
 r.s=0; r.se=0; k=0; ce=0; cs=0;
 /*@ loop invariant k >= 0 && k <= n && mathsum(k) == cs;
 @ loop invariant ((k % 2 == 0) ==> (mathse(k) == ce));
 @ loop invariant ((k % 2 != 0) ==> (mathse(k) == ce));
 loop assigns k, ce, cs;
 loop variant n-k;
 */
 while (k < n)
 {
 if (k % 2 != 0)
 { ce = ce + k + 1; cs = cs + k + 1; k = k + 1; }
 else
 { ce = ce ; cs = cs + k + 1; k = k + 1; }
 }
 r.s=cs; r.se=ce;
 return(r);
}
```

# Generation of an algorithm from M14

## Improving and checking with Frama-c

```
#include "structure.h"
#include "trans-even.h"
struct sums codesum(int n)
{
 int k, ce, cs; struct sums r;
 r.s=0; r.se=0; k=0; ce=0; cs=0;
 /*@ loop invariant k >= 0 && k <= n && mathsum(k) == cs;
 @ loop invariant ((k % 2 == 0) ==> (mathse(k) == ce));
 @ loop invariant ((k % 2 != 0) ==> (mathse(k) == ce));
 loop assigns k, ce, cs;
 loop variant n-k;
 */
 while (k < n)
 {
 if (k % 2 != 0)
 { ce = ce + k + 1; }
 cs = cs + k + 1; k = k + 1;
 }
 r.s=cs; r.se=ce;
 return(r);
}
```

# Generation of an algorithm from M14

Using the Plugin EB2Algo

```
rs : $\in \mathbb{Z}$ ||
re : $\in \mathbb{Z}$ ||
input = n ||
cur = 0 ||
ce = 0 ||
cs = 0

while cur $\neq$ n do
 if cur mod 2 $\neq$ 0 then
 cur = cur+1 ||
 cs = cs+cur+1 ||
 ce = ce+cur+1
 else
 cur = cur+1 ||
 cs = cs+cur+1 ||
 ce = ce
```

## Current Sub Section Summary

- 1 CM5
- 2 Overview
- 3 Programming by contract
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## Merging events

```

EVENT e1
ANY
 x
WHERE
 $G1(s, c, x, u)$
THEN
 $u : |(R(s, c, x, u, u'))|$
END

```

```

EVENT e2
ANY
 x
WHERE
 $G2(s, c, x, u)$
THEN
 $u : |(R(s, c, x, u, u'))|$
END

```

# Merging events

```
EVENT e1
ANY
 x
WHERE
 G1(s, c, x, u)
THEN
 u : |(R(s, c, x, u, u'))
END
```

```
EVENT e2
ANY
 x
WHERE
 G2(s, c, x, u)
THEN
 u : |(R(s, c, x, u, u'))
END
```

```
EVENT e refines e1, e2
ANY
 x
WHERE
 H(s, c, x, u)
THEN
 u : |(R(s, c, x, u, u'))
END
```

# Merging events

```
EVENT e1
ANY
 x
WHERE
 $G1(s, c, x, u)$
THEN
 $u : |(R(s, c, x, u, u'))$
END
```

```
EVENT e2
ANY
 x
WHERE
 $G2(s, c, x, u)$
THEN
 $u : |(R(s, c, x, u, u'))$
END
```

```
EVENT e refines e1, e2
ANY
 x
WHERE
 $H(s, c, x, u)$
THEN
 $u : |(R(s, c, x, u, u'))$
END
```

Checking the Proof Obligation :

$$Ax(s, c), I(s, c, u), H(s, c, x, u) \vdash \\ G1(s, c, x, u) \vee G2(s, c, x, u)$$

- Preparing merging by transforming actions.

**EVENT** e

**ANY**

$x$

**WHERE**

$G(s, c, x, u)$

**THEN**

$u : |(R(s, c, x, u, u'))$

**END**

**EVENT** f refines e

**ANY**

$x$

**WHERE**

$G(s, c, x, u)$

**THEN**

$u : |(G(s, c, x, u) \Rightarrow R(s, c, x, u, u'))$

**END**

- Internalizing the action as a before after relation
- $G(s, c, x, u) \wedge (G(s, c, x, u) \Rightarrow R(s, c, x, u, u')) \Leftrightarrow$   
 $G(s, c, x, u) \wedge R(s, c, x, u, u')$

- Preparing merging by transforming actions.

**EVENT** e

**ANY**

*x*

**WHERE**

$G(s, c, x, u)$

**THEN**

$u : |(R(s, c, x, u, u'))$

**END**

**EVENT** f refines e

**ANY**

*x*

**WHERE**

$G(s, c, x, u)$

**THEN**

$u : |(G(s, c, x, u) \Rightarrow R(s, c, x, u, u'))$

**END**

- $G(s, c, x, u) \wedge (G(s, c, x, u) \Rightarrow R(s, c, x, u, u')) \Leftrightarrow G(s, c, x, u) \wedge R(s, c, x, u, u')$
- $A(s, c, u) \wedge I(s, c, u) \wedge BA(f)(s, c, u, u') \Rightarrow I(s, c, u') \wedge BA(e)(s, c, u, u')$

- Preparing merging by transforming actions.

**EVENT** e

**ANY**

*x*

**WHERE**

$G(s, c, x, u)$

**THEN**

$u : |(R(s, c, x, u, u'))$

**END**

**EVENT** f refines e

**ANY**

*x*

**WHERE**

$G(s, c, x, u)$

**THEN**

$u : |(G(s, c, x, u) \Rightarrow R(s, c, x, u, u'))$

**END**

- $G(s, c, x, u) \wedge (G(s, c, x, u) \Rightarrow R(s, c, x, u, u')) \Leftrightarrow$   
 $G(s, c, x, u) \wedge R(s, c, x, u, u')$
- $A(s, c, u) \wedge I(s, c, u) \wedge BA(f)(s, c, u, u') \Rightarrow$   
 $I(s, c, u') \wedge BA(e)(s, c, u, u')$
- $A(s, c, u) \wedge I(s, c, u) \wedge G(s, c, x, u) \wedge (G(s, c, x, u) \Rightarrow$   
 $R(s, c, x, u, u')) \Rightarrow I(s, c, u') \wedge G(s, c, x, u) \wedge R(s, c, x, u, u')$

## Synthesis Phase : review of machines from M15-M20

**MACHINE** *M15* **SEES** *C0, C1*  
**VARIABLES**  
*input, rs, re, cur, cs, ce, l,*

**MACHINE** *M16* **SEES** *C0, C1*  
**VARIABLES**  
*input, rs, re, cur, cs, ce, l,*

## Synthesis Phase : review of machines from M15-M20

**MACHINE M17 SEES C0, C1**  
**VARIABLES**  
*input, rs, re, cur, cs, ce, l,*

**MACHINE** *M18* **SEES** *C0, C1*  
**VARIABLES**  
*input, rs, re, cur, cs, ce, l,*

**MACHINE** *M19* **SEES** *C0, C1*  
**VARIABLES**  
*input, rs, re, cur, cs, ce, l,*

## Synthesis Phase : review of machines from M15-M20

**MACHINE** *M20* **SEES** *C0, C1*  
**VARIABLES**  
*input, rs, re, cur, cs, ce, l,*

# Generating a TLA specification

```

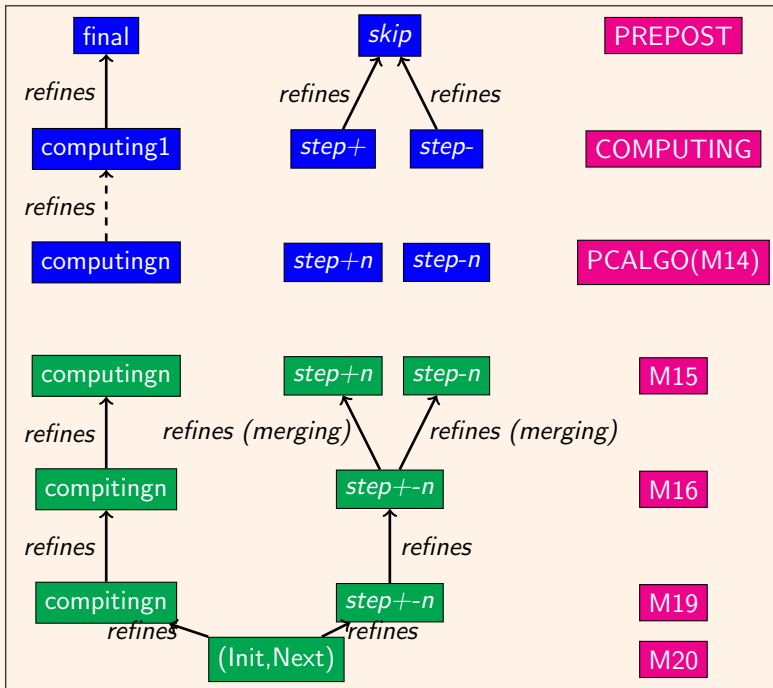
next1 ==
 (l="start" /\ cur < n /\ (cur % 2 = 0))
 /\ (cur'=cur+1 /\ ce'=ce /\ l'=l
 /\ cs'=cs+cur+1 /\ re'=re /\ rs'=rs)
o
delsnext2 ==
 (l="start" /\ cur < n /\ (cur % 2 # 0))
 /\ (cur'=cur+1 /\ ce'=ce+cur+1 /\ l'=l
 /\ cs'=cs+cur+1 /\ re'=re /\ rs'=rs)

next3 ==
 (l="start" /\ cur=n) /\ (rs'=cs /\ re'=ce
 /\ l'="end" /\ cur'=cur /\ cs'=cs /\ ce'=ce)

Next ==
 (next1 \/ next2 \/ next3)

Init == l="start" /\ cur=0 /\ rs=0 /\ cs=0 /\ re=0 /\ ce=0
=====

```



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- ⑤ Internalizing events by merging
- ⑥ The Call-as-Event Paradigm
- ⑦ Analysis and then Synthesis
  - Analysis using Refinement
  - Synthesis by Merging
- ⑧ Summing a list of integer values

# Contract for the problem Summing a list of integer values

Contract for the problem Summing  
a list of integer values

**VARIABLES**  $n, v, r, done$

---

**REQUIRES**  $\left\{ \begin{array}{l} n_0 \in \mathbb{N} \\ n_0 \neq 0 \\ v_0 \in 1..n_0 \rightarrow \mathbb{Z} \\ r_0 \in \mathbb{Z} \\ done_0 \in \text{BOOL} \end{array} \right.$

**ENSURES**  $\left( \begin{array}{l} r_f = \sum_{k=1}^{n_0} v_0(k) \\ done_f = \text{TRUE} \end{array} \right)$

---

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- the initial values of  $n$ ,  $v$  and  $done$ , denoted  $n_0$ ,  $v_0$ ,  $done_0$  and the final value of  $r$  denoted  $r_f$ .
- the final value of the variable  $r$  denoted  $r_f$  is the sum  $\sum_{k=1}^{n_0} v_0(k)$ .

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- the final value of the variable  $r$  denoted  $r_f$  is the sum  $\sum_{k=1}^{n_0} v_0(k)$ .

## computation method

Defining the computation process by defining a sequence  $u$  in the context  $S_0$

```
CONTEXT S0
CONSTANTS n0 v0 u
AXIOMS
 @pre1 n0 ∈ ℕ
 @pre2 n0 ≠ 0
 @pre5 v0 ∈ 1..n0 → ℤ
 @axm1 u ∈ 0..n0 → ℤ
 @axm2 u(0) = 0
 @axm3 ∀ k. k ∈ 1..n0
 ⇒ u(k) = u(k - 1) + v0(k)
end
```

```
CONTEXT S0
CONSTANTS n0 v0 u
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end
```

- The context S0 defines the sequence  $u$  supporting the computation method by definition of the summation expression.

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CONSTANTS n0 v0 u
AXIOMS
 @pre1 n0 ∈ ℕ
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 @pre5 v0 ∈ 1..n0 → ℤ
 @axm1 u ∈ 0..n0 → ℤ
 @axm2 u(0) = 0
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 ⇒ u(k) = u(k - 1) + v0(k)
end
```

- The context S0 defines the sequence  $u$  supporting the computation method by definition of the summation expression.
- The refinement process starts by the specification namely S1 and leads to the machine S7 which is containing the event final internalizing the computation of the sum.

# Expressing pre/post in a machine S1 (I)

```
MACHINE S1 SEES S0
VARIABLES r v n done
INVARIANTS
 @inv1 r ∈ ℤ
 @inv2 n ∈ ℤ
 @inv3 v ∈ 1..n → ℤ
 @inv4 done ∈ BOOL
 @read-values n = n0 ∧ v = v0
 @post done = TRUE
 ⇒ r = u(n0)
EVENT INITIALISATION
 then
 @act1 r : ∈ ℤ
 @act2 n := n0
 @act3 v := v0
 @act4 done := FALSE
 end
```

- $v$  and  $n$  are variables getting the input values as  $v0$  and  $n0$  : *read-values* label.
- $r$  is getting the result of the computing process under the control of *done* : post label.
- *post* : expression of the partial correctness.

## Expressing pre/post in a machine S1 (II)

- The final1 event expresses the relationship between initial and final values of variables.
- The keep event expresses the *hidden* loop preparing the final result.

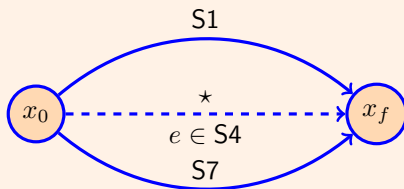
```

EVENT final1
 where
 @grd1 done = FALSE
 then
 @act1 r := u(n)
 @act2 done := TRUE
 end
anticipated EVENT keep
 then
 @act1 r, n, v, done :|
 (r' ∈ ℤ ∧ n' ∈ ℤ
 ∧ v' ∈ 1..n' → ℤ
 ∧ done' ∈ BOOL
 ∧ n' = n0 ∧ v' = v0
 ∧ (done' = TRUE
 ⇒ r' = u(n0)))
 end

```

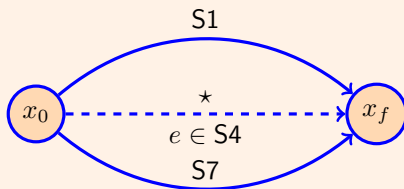
# Summary of the refinement steps

Refinement from S1  
till S7.



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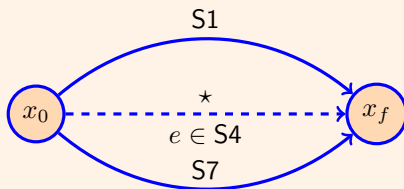
Refinement from S1  
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- Identifying the final1 event as simulating the pre/post contract in S1.

# Summary of the refinement steps

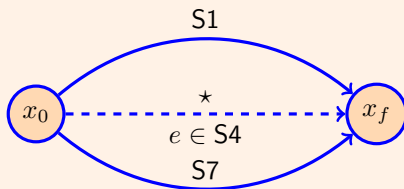
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- Identifying the final event as simulating the pre/post contract in S1.
- Refining by using  $u$  and introducing events simulating computing steps in S4.

# Summary of the refinement steps

Refinement from S1  
till S7.



- Identifying the final event as simulating the pre/post contract in S1.
- Refining by using  $u$  and introducing events simulating computing steps in S4.
- Merging events of S4 into one event in S7.

## From S1 to S4 via S2 and S3

## From S1 to S4 via S2 and S3

```

MACHINE S2 REFINES S1 SEES S0
VARIABLES r i uu n v done
INVARIANTS
 @inv1 $i \in 0..n$
 @inv2 $uu \in 0..n \mapsto \mathbb{Z} \wedge \text{dom}(uu) = 0..i$
 @inv3 $\forall k. k \in \text{dom}(uu) \Rightarrow uu(k) = u(k)$
 @inv4 $\text{done} = \text{FALSE} \Rightarrow i \in 0..n$
 @inv6 $\text{done} = \text{TRUE} \Rightarrow i = n$
 @inv7 $i \in 0..n - 1 \Rightarrow \text{done} = \text{FALSE}$
theorem @datath $v = v0 \wedge n = n0$
VARIANT $n - i$

```

- The variable  $uu$  is recording the sequence  $u$  by computing until  $i$  is  $n0$ .
- The new event step is refining the event keep and decreases the variant.

# From S1 to S4 via S2 and S3

```

MACHINE S2 REFINES S1 SEES S0
VARIABLES r i uu n v done
INVARIANTS
 @inv1 $i \in 0..n$
 @inv2 $uu \in 0..n \rightarrow \mathbb{Z} \wedge \text{dom}(uu) = 0..i$
 @inv3 $\forall k. k \in \text{dom}(uu) \Rightarrow uu(k) = u(k)$
 @inv4 $\text{done} = \text{FALSE} \Rightarrow i \in 0..n$
 @inv6 $\text{done} = \text{TRUE} \Rightarrow i = n$
 @inv7 $i \in 0..n - 1 \Rightarrow \text{done} = \text{FALSE}$
theorem @datath $v = v0 \wedge n = n0$
VARIANT $n - i$

```

```

MACHINE S3 REFINES S2 SEES S0
VARIABLES r i uu cu n v done
INVARIANTS
 @inv1 cu ∈ ℤ
 @inv2 cu = u(i)

```

- The variable  $uu$  is recording the sequence  $u$  by computing until  $i$  is  $n0$ .
- The new event step is refining the event `keep` and decreases the variant.
- The new variable  $cu$  is recording the last computed value of  $uu$ .
- Proofs of the invariants are made easier.

# The machine S4 in the middle (I)

```
MACHINE S4 REFINES S3 SEES S0
VARIABLES r i cu n v done
INVARIANTS
 theorem @inv1 cu = u(i)
EVENTS
 EVENT INITIALISATION
 then
 @act1 r : $\in \mathbb{Z}$
 @act2 i := 0
 @act4 cu := 0
 @act5 v := v0
 @act6 n := n0
 @act7 done := FALSE
 end
```

## The machine S4 in the middle (I)

```

MACHINE S4 REFINES S3 SEES S0
VARIABLES r i cu n v done
INVARIANTS
 theorem @inv1 cu = u(i)
EVENTS
 EVENT INITIALISATION
 then
 @act1 r : $\in \mathbb{Z}$
 @act2 i := 0
 @act4 cu := 0
 @act5 v := v0
 @act6 n := n0
 @act7 done := FALSE
 end

```

- The variable `uu` helps in the proof process but is expensive.
- The variable `uu` is hidden in the abstractions.

# The machine S4 in the middle (II)

```
EVENT final REFINES final
 where
 @grd1 i = n
 @grd2 done = FALSE
 then
 @act1 r := cu
 @act2 done := TRUE
 end

EVENT step000 REFINES step
 where
 @grd1 i < n
 @grd2 done = FALSE
 then
 @act1 i := i + 1
 @act3 cu := cu + v(i + 1)
 end
```

## The machine S4 in the middle (II)

```

EVENT final REFINES final
 where
 @grd1 $i = n$
 @grd2 $done = FALSE$
 then
 @act1 $r := cu$
 @act2 $done := TRUE$
 end
end

```

- Events are made simpler and  $uu$  is hidden.
- Events are guarded by evaluable conditions as  $i < n$  which is implying  $i \notin \text{dom}(uu)$
- Using EB2ALGO, one gets an algorithm but no proof is given of the transformations.

# The machine S7 with one event

```
MACHINE S7 REFINES S6 SEES S0
VARIABLES r i cu n v done
INVARIANTS
 theorem @inv1 done = FALSE $\Rightarrow$ $i \in 0..n$
 theorem @inv2 done = TRUE $\Rightarrow$ $i = n$
 theorem @inv3 $i \in 0..n - 1 \Rightarrow$ done = FALSE
 theorem @inv4 $i \in 0..n$
EVENT final REFINES final
 where
 @grd2 done = FALSE
 then
 @act1 r, i, cu, n, v, done :| (
 (i = n
 $\Rightarrow$ ($r' = cu \wedge cu' = cu \wedge i' = i \wedge n' = n$
 $\wedge v' = v \wedge done' = TRUE$))
 $\wedge (i < n$
 $\Rightarrow$ ($r' = r \wedge cu' = cu + v(i + 1) \wedge i' = i + 1$
 $\wedge n' = n \wedge v' = v \wedge done' = done$))
)
 end
```

## Defining an algorithm by a set of actions

## Defining an algorithm by a set of actions

$$guard(r, i, cu, n, v, done) \equiv done = FALSE$$

$$action \equiv \begin{cases} i = n \Rightarrow \begin{cases} (r' = cu \wedge cu' = cu \wedge i' = i \wedge n' = n \\ \wedge v' = v \wedge done' = TRUE) \\ r' = r \wedge cu' = cu + v(i+1) \wedge i' = i+1 \wedge n' = n \\ \wedge v' = v \wedge done' = done \end{cases} \\ i < n \Rightarrow \end{cases}$$

## Defining an algorithm by a set of actions

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- The machine  $S7$  contains an event Initialisation and a next relation
-

## Defining an algorithm by a set of actions

$$\begin{aligned} guard(r, i, cu, n, v, done) &\equiv done = FALSE \\ action &\equiv \begin{cases} i = n \Rightarrow \begin{cases} (r' = cu \wedge cu' = cu \wedge i' = i \wedge n' = n \\ \wedge v' = v \wedge done' = TRUE) \end{cases} \\ i < n \Rightarrow \begin{cases} r' = r \wedge cu' = cu + v(i+1) \wedge i' = i+1 \wedge n' = n \\ \wedge v' = v \wedge done' = done \end{cases} \end{cases} \end{aligned}$$

- The machine  $S7$  contains an event Initialisation and a next relation
- 

## Definition

(mapping of a binary relation)

A binary relation  $R$  on  $State$  can be mapped to a GKAT term  $e$  for the interpretation  $i$ , if  $\mathcal{R}_i(e) = R$  and we write  $map_i(R) = e$



## Defining an algorithm by a set of actions (II)

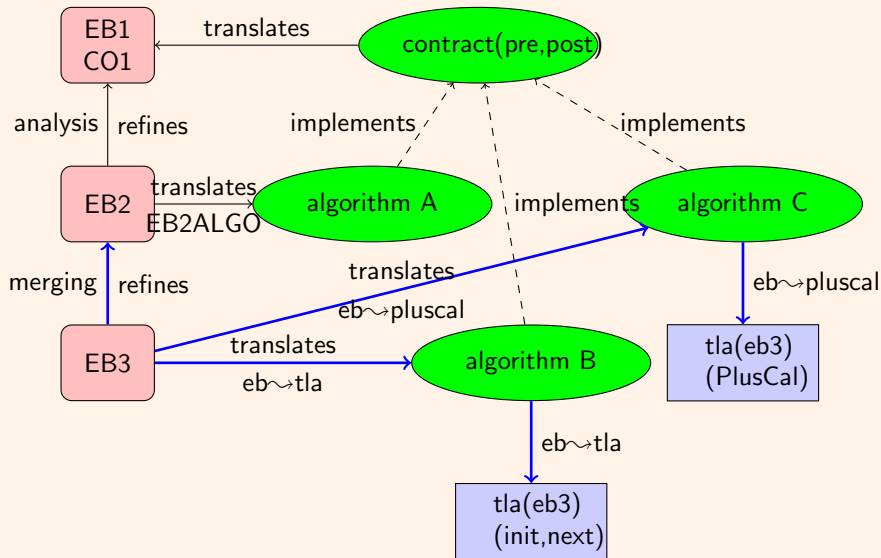
### Listing 4 – PlusCal algorithm for summing a vector

```

—algorithm sumvect {
variables n=n0, v=v0, i=0, cu=0, r;
{
w1: while (i $\neq$ n) {
 w2: cu:=cu+v[i +1]; i:=i+1;
};
w3: r:=cu; Gu
}}

```

# Global View



- Algorithm A is generated using EB2Algo

$$\begin{array}{l} r := \mathbb{Z} \parallel i := 0 \parallel cu := 0 \parallel v := v0 \parallel n := n0 \\ \text{while } i < n \text{ do} \\ \quad i := i + 1 \parallel cu := cu + v(i + 1) \\ \text{od;} \end{array}$$

- Algorithm B is simply defined by

$$guard(r, i, cu, n, v, done) \equiv done = FALSE$$

$$action \equiv \begin{cases} i = n \Rightarrow \begin{cases} (r' = cu \wedge cu' = cu \wedge i' = i \wedge n' = n \\ \wedge v' = v \wedge done' = TRUE) \end{cases} \\ i < n \Rightarrow \begin{cases} r' = r \wedge cu' = cu + v(i+1) \wedge i' = i+1 \wedge n' = n \\ \wedge v' = v \wedge done' = done \end{cases} \end{cases}$$

Algorithm C is generated using our transformations based on GKAT algebra

```
--algorithm sumvect {
variables n=n0,v=v0,i=0,cu=0,r;
{
w1:while (i # n) {
 w2:cu:=cu+v[i+1]; i:=i+1;
};
w3:r:=cu;
}}
```